

Isaac Mubiru

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PROFESSIONAL SUMMARY

Senior backend engineer with 5+ years on Node.js and TypeScript SaaS systems — built developer-facing reliability tooling, real-time data pipelines, and zero-downtime migrations that other engineers depend on. Code lives at github.com/zac-09.

Led the migration of 20+ production apps from Parse/MongoDB to Firebase at MTailor with zero downtime, saving ~\$5K/month in infrastructure cost — exactly the customer pain that monitoring platforms solve. Strong distributed-systems foundation (Apache Kafka + NATS Streaming + gRPC microservices, 4 NGO tenants on shared infra at CodeBits) and async-first remote experience: 5+ years on US-based teams from Uganda (UTC+3, edge of your stated window, full EMEA overlap).

CORE COMPETENCIES

Node.js

TypeScript

SaaS Backend

SQL (Postgres-ready)

Apache Kafka

NATS Streaming

gRPC

Pub/Sub Pipelines

AWS / GCP

Docker / Kubernetes

Playwright

Zero-Downtime Migrations

Async-First Remote

On-Call / Incident Response

WORK EXPERIENCE

MTailor Sep 2022 – Present

Software Engineer · US Remote (Uganda, UTC+3)

- Led the end-to-end migration of **20+ production applications** from Parse/MongoDB to Firebase/GCP with **zero downtime**, reporting directly to the CTO.
- Implemented **real-time two-way sync between MongoDB and Firestore** using Node.js and Google Pub/Sub — observable parallel write path with drift-detection reconciliation jobs.
- Saved **~\$5,000/month** by migrating off AWS — audited services, migrated S3→GCS via Python, lifted EC2 workloads to Cloud Functions where viable.
- Built a 3D visualization feature using video overlay and ffmpeg, increasing buyer conversion.
- Wrote internal SDK guides and trained the Ops team on Firebase Dashboard + backup procedures.

Dr Wealth Aug 2021 – Jan 2022

Software Engineer · Singapore Remote

- Led optimization of a PWA delivering real-time stock prices from a Firebase backend.
- Built ad-hoc analytical jobs querying **2M+ Firestore records**, keeping prices current from Morningstar APIs with sub-15-min freshness.
- Extended backend APIs using Firebase Cloud Functions and Express.js on Heroku.

CodeBits Jan 2020 – Jul 2021

Software Engineer · Lead of 4 · Uganda Remote

- Architected a **microservices backend with Apache Kafka, NATS Streaming, gRPC, Docker, and Kubernetes** for FIDA Uganda — independent service deploys, 4 NGO tenants on shared infra.
- Built a USSD legal-aid service over flaky 2G — Node.js backend with retry-aware session state machine, instrumented per-region latency.
- Built a cross-platform mobile app for LASPNET using React Native + Expo with Firebase push.

Mind2matter.co Jan 2022 – Jul 2022

Software Engineer · US Remote

- Built backends for DeFi applications using Web3 + Node.js — contract integration, transaction signing, off-chain indexing.
- Delivered React UIs from Figma designs under tight client timelines.

EDUCATION

B.Sc. Computer Engineering — **Makerere University** Aug 2018 – Sep 2022

SKILLS

Proficient: Node.js, TypeScript, JavaScript, SQL, MongoDB, Firebase/Firestore, Apache Kafka, NATS Streaming, gRPC, GCP, Python, Docker, Kubernetes, Linux, Playwright

Working knowledge: Postgres (SQL fundamentals), AWS, ClickHouse-curious (analytical-scale Firestore work as proxy)

Intermediate: Go, Rust, React Native, Web3

Languages: English (native-level), Kiswahili